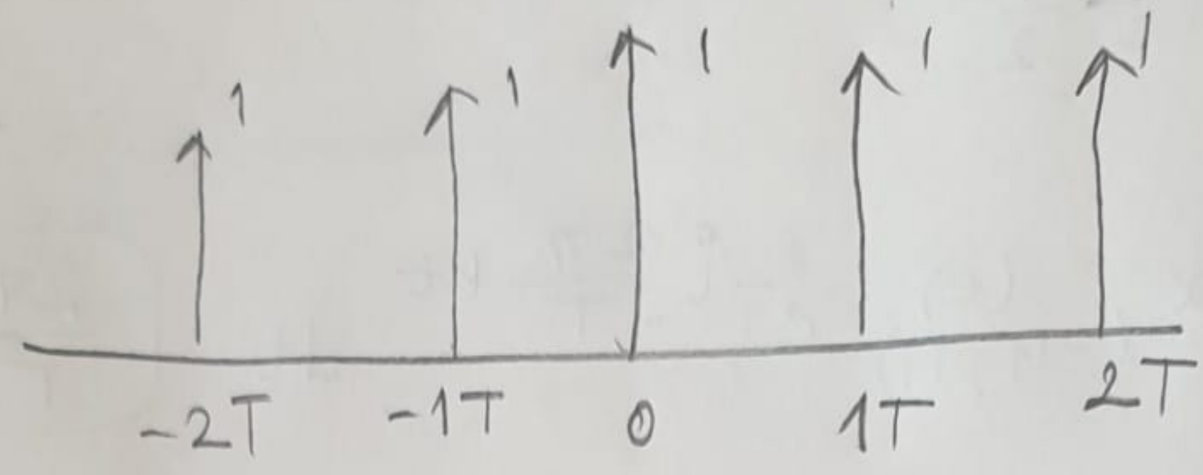
(ex) $FT\left[\sum_{k=-\infty}^{\infty} \delta(t-Tk)\right]$



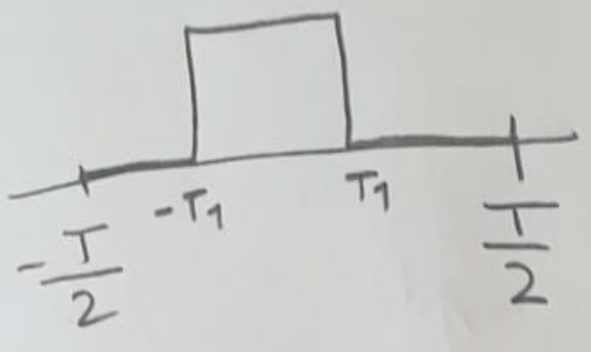
$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\, e^{-jk\omega t}\, dt = \frac{1}{T}$$

$$\underset{x(w)}{FT} = \frac{2\pi}{T}\sum \delta\left(w - \frac{2\pi}{T}k\right)$$

Prop of FiT cnt.

$-\delta(\omega-\alpha)]$

(ex)



FT find

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x_{(-T_1,T_1)}^{(t)}\, e^{-i\frac{2\pi}{T}kt}\, dt \qquad \boxed{\frac{2\pi}{T}=\omega_0}$$

$$= \frac{1}{T}\int_{-T_1}^{T_1} x_{(-T_1,T_1)}^{(t)}\, e^{-i\omega_0 kt}\, dt$$

$$\frac{1}{T}\left[\frac{e^{-i\omega_0 kt}}{-i\omega_0 k}\right|_{-T_1}^{T_1} = \frac{e^{-i\omega_0 kT_1}}{-i\omega_0 k} + \frac{e^{i\omega_0 kT_1}}{+i\omega_0 k}$$

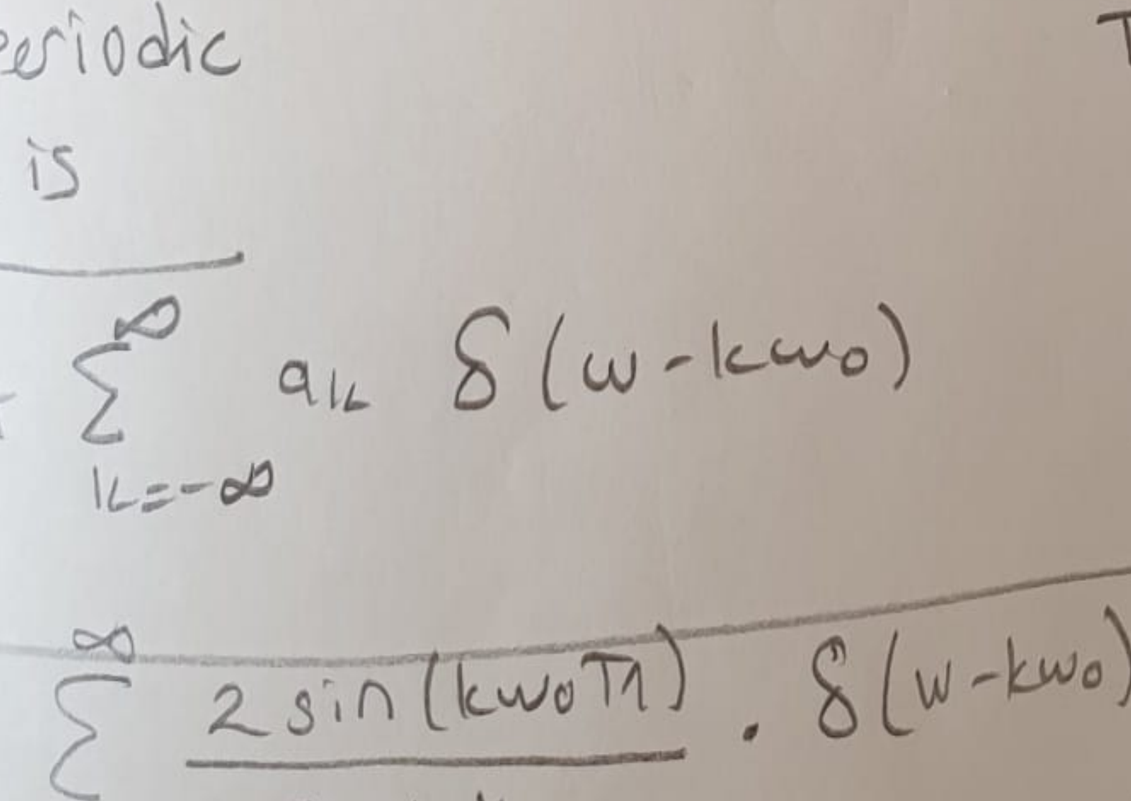
$$= 2\cdot\frac{e^{i\omega_0 kT_1} - e^{-i\omega_0 kT_1}}{2\cdot i\omega_0 k}\cdot\frac{1}{T}$$

$$= \frac{2\sin(k\omega_0 T_1)}{T\ \omega_0 k}$$

FT of periodic thing is

$$2\pi\sum_{k=-\infty}^{\infty} a_k\,\delta(\omega-k\omega_0)$$

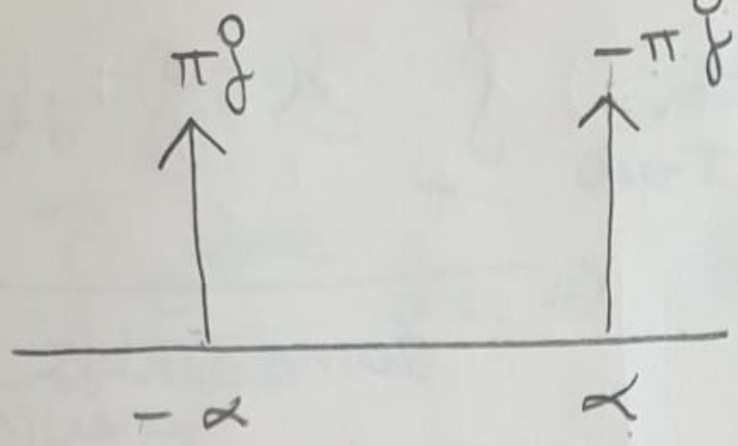
$$= 2\pi\sum_{k=-\infty}^{\infty}\frac{2\sin(k\omega_0 T_1)}{T\omega_0 k}\cdot\delta(\omega-k\omega_0)$$



$(\omega+\alpha)$

$+j\sin(-\omega t)$

$\pi j [\delta(w+\alpha) - \delta(w-\alpha)]$



ex

$FT[\cos(\alpha t)]$

$$\int_{-\infty}^{\infty} \frac{e^{\alpha jt} + e^{-\alpha jt}}{2} \cdot e^{-jwt}\, dt$$

$e^{jt(\alpha-w)}$ $e^{jt(\alpha+w)}$

$$\frac{\pi\delta(w-\alpha) + \pi(w+\alpha)}{2} = \pi\delta(w-\alpha) + \pi\delta(w+\alpha)$$

ex

$FT[u(t)]$

δ_0

$$\int_{-\infty}^{\infty} u(t) \cdot e^{-jwt}\, dt = \int_{0}^{\infty} e^{-jwt}\, dt$$

$$= \lim_{T\to\infty} \frac{e^{-jwt}}{-jw}\Big|_0^T = \lim_{T\to\infty} \frac{e^{-jwt}}{-jw} + \frac{1}{+jw}$$

$$= \frac{1}{jw} - \frac{\lim_{T\to\infty} e^{-jwt}}{jw} \rightarrow \cos(-wT) + j\sin(-wT)$$

$$= \frac{1}{jw} - \lim_{T\to\infty} \left[\frac{\cos(jwT)}{jw} - \frac{j\sin(jwT)}{jw} \right]$$

$$= \frac{1}{jw} + \lim_{T\to\infty} \frac{\sin(wT)}{w} \Big/ jw \quad (0)$$

$$= \frac{1}{jw} + \pi\delta(w)$$

ex

$ak =$

$=$

FT of p
thing

$= 2\pi$

$$x(w) = \int_{-\infty}^{\infty} e^{j\alpha t} \cdot e^{-jwt}\, dt \Rightarrow \lim_{T\to\infty} \int_{-T}^{T} e^{j(\alpha-w)t}\, dt$$

same with previous one.

previous was

$$\lim_{\alpha\to\infty} \int_{-a}^{a} e^{-jwt} \Rightarrow 2\pi\,\delta(w)$$

this will be

$$\lim_{T\to\infty} \int_{-T}^{T} e^{-j(\alpha+w)t}\, dt \Rightarrow \underline{2\pi\,\delta(w-\alpha) \overset{or}{=} 2\pi\,\delta(\alpha-w)}$$

$$= \lim_{T\to\infty} 2\,\frac{\sin[(\alpha-w)T]}{\alpha-w} = \nearrow \bigstar$$

(ex) FT $[\sin(\alpha t)]$

$$\int_{-\infty}^{\infty} \sin(\alpha t) \cdot e^{-jwt}\, dt$$

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i} \quad (i=j) \Rightarrow \sin(\alpha t) = \frac{e^{j\alpha t} - e^{-j\alpha t}}{2j}$$

$$FT\left[\frac{e^{j\alpha t} - e^{-j\alpha t}}{2j}\right] = \frac{FT(e^{j\alpha t}) - FT(e^{-j\alpha t})}{2j}$$

$\star\star\star$ $\quad$ $\star\star\star$

$$\frac{2\pi\,\delta(\alpha-w) - 2\pi\,\delta(\alpha+w)}{2j} = \pi\,\frac{1}{j}\left[\delta(\alpha+w) - \delta(\alpha-w)\right]$$

$$\boxed{\frac{1}{j} = -j}$$

$$\boxed{v_1 \cdot v_2 = |v_1| \cdot |v_2| \cdot \cos\theta}$$

11.05. Signal

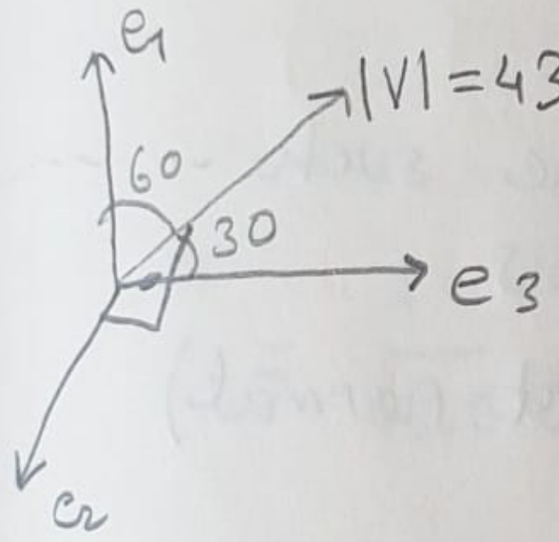
Basis Thm

$e_i \cdot e_j = \delta_{ij}$

if $i \neq j$ they are perpendicular.

$$\boxed{e_m \cdot v = \alpha_m}$$

$$v = (v \cdot e_1) \cdot e_1 + (v \cdot e_2) \cdot e_2 + \cdots$$



$$v = (v \cdot e_1) \cdot e_1 + (v \cdot e_2) \cdot e_2 + (v \cdot e_3) \cdot e_3$$

$$v \cdot e_1 = 43 \cdot 1 \cdot \cos 60 = 43/2$$
$$v \cdot e_2 = 43 \cdot 1 \cdot \cos 90 = 0$$
$$v \cdot e_3 = 43 \cdot 1 \cdot \cos 30 = 43\sqrt{3}/2$$

$$v = \frac{43 + 43\sqrt{3}}{2} = \frac{43}{2} \cdot e_1 + 0 e_2 + \frac{43\sqrt{3} e_3}{2}$$

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \Rightarrow v \cdot w = 4 + 10 + 18 = 32$$

in function spaces, dot pr. of two func. $f(t)$ $g(t)$

$$\langle f(t), g(t) \rangle = \int_0^L f(t)\, g^*(t)\, dt$$

$$|f(t)| = \sqrt{\langle f(t), f(t) \rangle} = \sqrt{\int_0^L f(t) \cdot f^*(t)\, dt}$$

angle btw. func.

$$\cos\theta = \frac{\langle f(t), g(t) \rangle}{|f(t)|\,|g(t)|}$$

(cosine similarity)

want to express this in a diff. orthonormal basis.

$e^{i\frac{2\pi}{L}kt}$ is one such basis.

(orthogonal, not normal)

Fourier Series

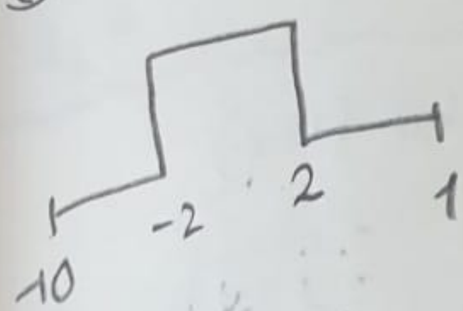
$$a_n = \frac{1}{L} \int_0^L f(t)\, e^{-i\frac{2\pi}{L}nt}\, dt$$

(L ← variable, 0 ← variable)

$$f(t) = \sum_{k=-\infty}^{\infty} a_k e^{i\frac{2\pi}{L}kt}$$

g(t)

(ex) example bizim işimiz.



$\Rightarrow \chi_{(-2,2)}(t)$

$$a_m = \frac{1}{20}\int_{-10}^{10} \chi_{(-2,2)}(t)\, e^{-i\frac{2\pi}{20}\cdot mt}\, dt$$

$$= \frac{1}{20}\int_{-2}^{2} \chi_{(-2,2)}(t)\, e^{-i\frac{\pi}{10}mt}\, dt$$

1 in that int.

$$= \frac{1}{20}\int_{-2}^{2} e^{-i\frac{\pi}{10}mt}\, dt$$

$$= \frac{1}{20}\left(\frac{e^{-i\frac{\pi}{10}mt}}{-i\cdot\frac{\pi}{10}\cdot m}\Bigg|_{-2}^{2}\right) = \frac{1}{20}\left(\frac{e^{-i\frac{\pi}{5}m} - e^{i\frac{\pi}{5}m}}{-i\frac{\pi}{10}m}\right)$$

$$= \frac{-e^{-i\frac{\pi}{5}m} + e^{i\frac{\pi}{5}m}}{2\cdot(+i\pi m)}$$

$$\cos x = \mathrm{Re}(e^{ix}) = \frac{e^{ix}+e^{-ix}}{2}$$

$$\sin x = \mathrm{Im}(e^{ix}) = \frac{e^{ix}-e^{-ix}}{2i}$$

$$a_m = \sin\left(\frac{\pi}{5}m\right)\cdot\frac{1}{\pi m}$$

$$\chi_{(-2,2)}(t) = \sum_{m=-\infty}^{\infty} \underbrace{\sin\left(\frac{\pi}{5}m\right)\cdot\frac{1}{\pi m}}_{a_m} \cdot \underbrace{e^{i\frac{2\pi}{20}mt}}_{\text{vector}}$$

ex

FT of $\delta(t)$

$$X(w) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-jwt}\, dt \Rightarrow \text{this lives only at } t=0 \quad \text{so} \quad = (1) \quad ??$$

ex

FT of $\delta(t-2)$

$$X(w) = \int_{-\infty}^{\infty} \delta(t-2) \cdot e^{-jwt}\, dt = e^{-jw2}$$

ex

FT of $e^{-2t}u(t)$

$$\int_{-\infty}^{\infty} e^{-2t}(u(t)) \cdot e^{-jwt}\, dt = \int_{0}^{\infty} e^{-2t}\, e^{-jwt}\, dt$$

$$= \int_{0}^{\infty} e^{-(2+jw)t}\, dt$$

$e^{-j\infty}$?

$$= \frac{e^{-(2+jw)t}}{-(2+jw)}\Bigg|_0^{\infty} = \frac{e^{-(2+jw)\infty} - 1}{(-2-jw)}$$

$$\hookrightarrow \frac{e^{-2\infty} - e^{-jw\infty}}{} = 0$$

$$= 0 - \frac{+1}{-2+jw}$$

$$= \frac{1}{2+jw}$$

ex

$$X(w) = \int_{-\infty}^{\infty}$$

instead

$\lim_{\tau}$

$$a_k = \frac{1}{L}\int_0^L f(t)\cdot e^{-i\frac{2\pi}{L}\cdot k\cdot t}\,dt$$

$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} 1\cdot e^{-i\frac{2\pi}{T}kt}\,dt$$

$$= \frac{1}{T}\left(\frac{e^{-i\frac{2\pi}{T}kt}}{-i\frac{2\pi}{T}k}\Bigg|_{-T_1}^{+T_1}\right) = \frac{1}{T}\left(\frac{e^{-i\frac{2\pi}{T}k\cdot T_1} - e^{+i\frac{2\pi}{T}kT_1}}{+i\frac{2\pi}{T}k}\right)$$

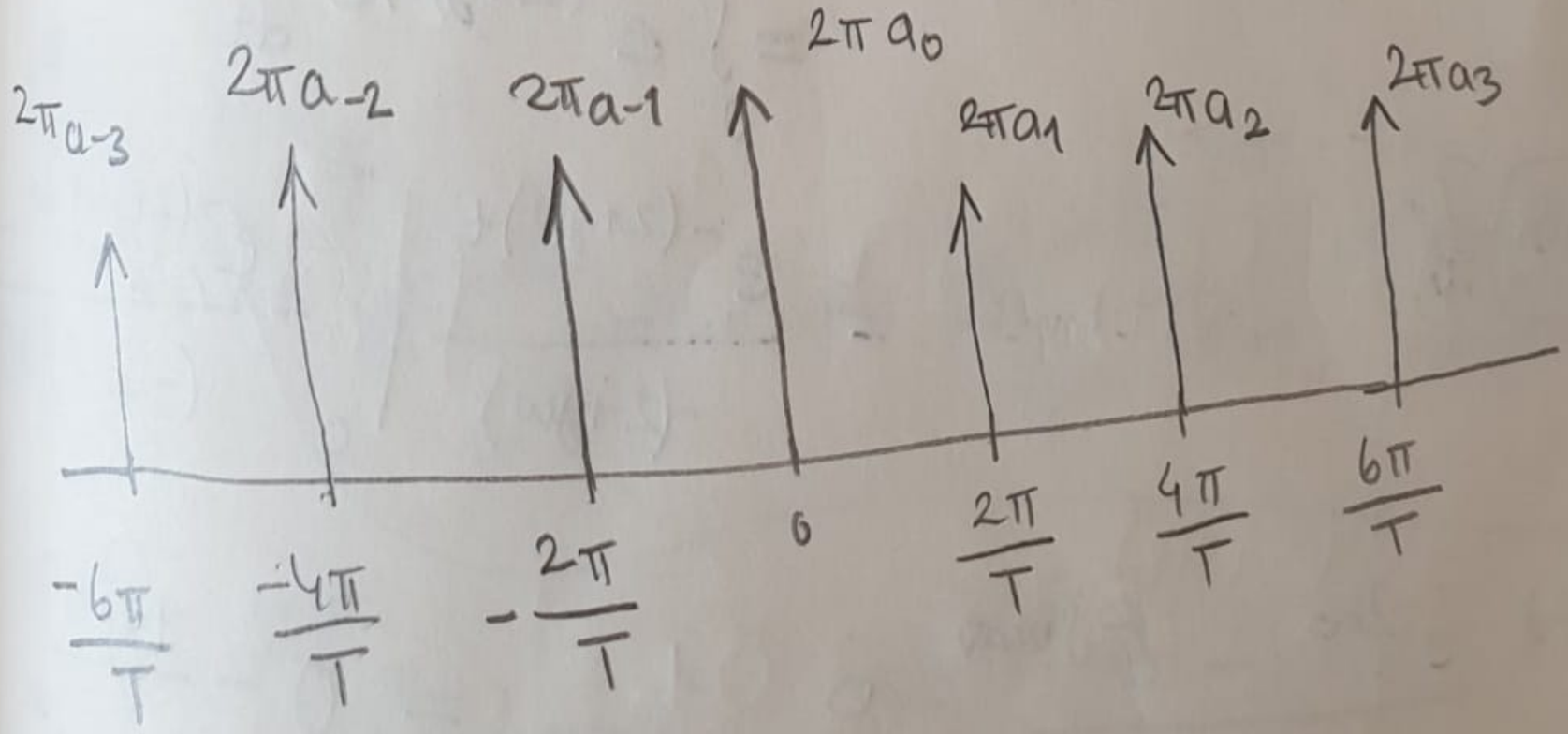
$$a_k = \frac{1}{\pi k}\sin\left(\frac{2\pi}{T}kT_1\right)$$

$$\propto \sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\propto \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

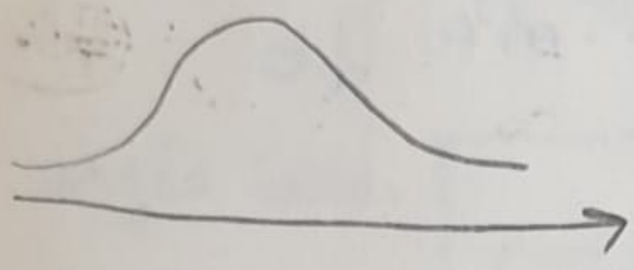
$$X(w) = 2\pi\sum_{k=-\infty}^{\infty} a_k\,\delta\left(w - \frac{2\pi k}{T}\right)$$

$$X(w) = 2\pi\sum_{k=-\infty}^{\infty}\frac{1}{\pi k}\sin\left(\frac{2\pi}{T}kT_1\right)\delta\left(w - \frac{2\pi k}{T}\right)$$



12.05. Signal

continues FS



$T \to \infty$

$$\frac{2\pi}{T} = \omega_0 \left(\frac{rad}{s}\right)$$

(time to freq.)

(FT)
$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

(IFT)
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

(freq. to time)

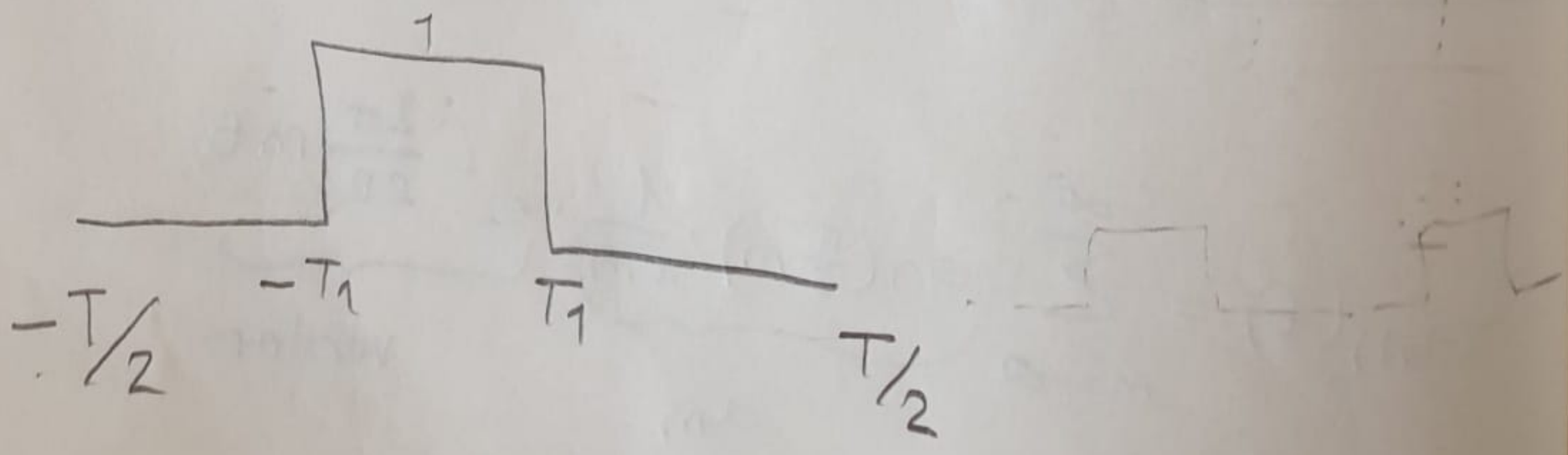
18.05. Signal

- FS is only Periodic and Intervaled func.
- FT is everything.

$$X(\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k\, \delta\left(\omega - \frac{2\pi k}{T}\right)$$

ex Find the F.T

first find a_k, then find $X(\omega)$.



ex

FT(1)

$$X(w) = \int_{-\infty}^{\infty} e^{-jwt}\, dt \Rightarrow \lim_{T\to\infty} \int_{-T}^{T} e^{-jwt}\, dt$$

$$= \frac{e^{-jwt}}{-jw}\Big|_{-T}^{T} = \frac{e^{-jwT}}{-jw} - \frac{e^{+jwT}}{-jw}$$

$$= \frac{e^{jwt} - e^{-jwt}}{-jw}$$

sdn

$$\lim_{\varepsilon\to 0} \frac{\sin(t/\varepsilon)}{t} = \pi\delta(t) = \lim_{a\to\infty} 2\frac{\sin(aw)}{w}$$

$w = 1/\varepsilon$ $\quad a = T$ $\quad = 2\pi\delta(w)$

we can write

ex

FT $(e^{j\alpha t})$

$$X(w) = \int_{-\infty}^{\infty} e^{jwt} \cdot e^{-jwt}\, dt = \int_{-\infty}^{\infty} 1\, dt = \infty = t\Big|_{-\infty}^{\infty} = \infty$$

25.05

ex) FT $\delta(t)$

$$x(w) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-jwt}\, dt =$$

integrali alduruyor
→ 0 layan değeri

yerine koy $\Rightarrow e^{-jw.0} = e^0 = 1$ //

ex) FT $[\delta(t-a)]$

$$x(w) = \int_{-\infty}^{\infty} \delta(t-a) \cdot e^{-jwt}\, dt$$

$\delta(t-a)$ → 1 ve → a

e^{-jwa} //

$\delta(w)$

ex) FT $[x_{(-a,a)}(t)]$

$$x(w) = \int_{-a}^{a} 1 \cdot e^{-jwt}\, dt = \frac{e^{-jwt}}{-jw}\Big|_{-a}^{a} = \frac{e^{-jwa} - e^{jwa}}{-jw}$$

$$= 2\left(\frac{e^{jwa} - e^{-jwa}}{2j}\right)\frac{1}{w} \qquad j = i$$

$$= \frac{2\sin(wa)}{w}$$ //

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$= 2 \lim_{\tau \to \infty} \frac{e^{-j\omega t} - e^{j\omega t}}{2 - j\omega} \qquad j = i$$

$$= 2 \lim_{T \to \infty} \frac{1}{\omega} \cdot \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

$$= 2 \lim_{T \to \infty} \frac{\sin(\omega T)}{\omega}$$

$$2 \cdot \lim_{\varepsilon \to 0} \frac{\sin(\omega t/\varepsilon)}{\omega}$$

$$= \underline{2\pi \delta(\omega)} \qquad FT(1) = \underline{\underline{2\pi \delta(\omega)}}$$

(ex) FT ex's

$$e^{-\alpha t} u(t) \qquad \alpha > 0$$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-\alpha t} u(t) \cdot e^{-j\omega t} dt = \int_{0}^{\infty} e^{-t(\alpha + j\omega)} dt$$

$$= \frac{e^{-t(\alpha + j\omega)}}{-(\alpha + j\omega)} \Bigg|_0^{\infty} = 0 + \frac{1}{\alpha + j\omega}$$

ex

$$e^{-2|t|}$$

$$X(w) = \int_{-\infty}^{\infty} e^{-jwt}\, dt$$

$$= \int_{-\infty}^{0} e^{2t} \cdot e^{-jwt}\, dt + \int_{0}^{\infty} e^{-2t} \cdot e^{-jwt}\, dt$$

$$= \frac{e^{t(2-jw)}}{2-jw}\Bigg|_{-\infty}^{0} + \frac{e^{-t(2+jw)}}{-2-jw}\Bigg|_{0}^{\infty}$$

$$= \frac{1}{2-jw} - 0 + 0 + \frac{1}{+2+jw}$$

$$= \frac{1}{2-jw} + \frac{1}{2+jw} = \frac{2}{4+w^2} //$$

ex FT(1)

$$X(w) = \int_{-\infty}^{\infty} \cdot e^{-jwt}\, dt = \frac{e^{-jwt}}{-jw}\Bigg|_{-\infty}^{\infty} = \text{Doesnot near anything}$$

instead do?

$$\lim_{T \to 0} \int_{-T}^{T} 1 \cdot e^{-jwt}\, dt$$

↓